

Class 12 Biology: High-Yield Diagram & Cross Checklist

Chapter 4: Principles of Inheritance and Variation (NCERT Page-wise Guide)

Instructions for Students: Practice genetic crosses, Punnett squares, and pedigree charts with clean pencil work. Always indicate gametes, phenotypic ratios, and genotypic ratios clearly.

Fig No.	Cross / Diagram Title	NCERT	Priority	Key Components / Ratios to Practice
Fig 4.4	Monohybrid Cross (Punnett Square)	Pg 57	★★★ Top Priority	$TT \times tt \rightarrow F_1 (Tt)$. F_2 Phenotypic (3:1), Genotypic ratio (1:2:1)
Fig 4.7	Dihybrid Cross (Seed Shape & Colour)	Pg 63	★★★ Top Priority	$RRYY \times rryy \rightarrow$ 16-box Punnett square. Phenotypic ratio: 9:3:3:1
Fig 4.6	Incomplete Dominance (Snapdragon)	Pg 60	★★★ Top Priority	RR (Red) $\times$ rr (White) $\rightarrow Rr$ (Pink). F_2 Phenotypic = Genotypic ratio (1:2:1)
Fig 4.5	Test Cross (Dominant Phenotype)	Pg 59	★★★ Top Priority	Crossing unknown violet flower with homozygous recessive (ww); 1:1 ratio if heterozygous
Fig 4.15	Sickle-Cell Anaemia (HbA vs HbS)	Pg 74	★★★ Top Priority	Point mutation: $GAG \rightarrow GUG$ codon; Glu $\rightarrow$ Val substitution at 6th position of β -globin chain
Fig 4.13	Pedigree Analysis Standard Symbols	Pg 72	★★★ Top Priority	Male ($\square$), Female ($\circ$), Affected (filled), Mating, Consanguineous mating
Fig 4.14	Pedigree Charts (Autosomal Traits)	Pg 73	★★☆ High Priority	(a) Autosomal Dominant (Myotonic dystrophy), (b) Autosomal Recessive (Sickle-cell)
Fig 4.13 (b)	Sex Determination in Honey Bee	Pg 71	★★☆ High Priority	Haplodiploidy: Female $2n=32$ (meiosis), Male $n=16$ (mitosis/parthenogenesis)
Fig 4.12	Sex Determination Systems	Pg 70	★★☆ High Priority	(a, b) Male heterogamety: XY/XX (Human, Drosophila), (c) Female heterogamety: ZW/ZZ (Birds)
Fig 4.11	Morgan's Linkage & Recombination	Pg 68	★★☆ High Priority	Cross A (y-w: 1.3% recomb - tight linkage), Cross B (w-m: 37.2% recomb - loose linkage)
Fig 4.16 & 4.17	Chromosomal Disorders Symptoms	Pg 76	★★☆ Moderate	Down's (Trisomy 21), Klinefelter's (47, XXY), Turner's (45, XO)

KEY BOARD EXAM DRAWING & SCORING GUIDELINES

- **Cross Formatting:** Clearly label Parents (P), Genotypes, Gametes, F_1 generation, Selfing, and F_2 Punnett Square in monohybrid/dihybrid crosses.
- **Molecular Label in Sickle-Cell (Fig 4.15):** Memorize the precise base triplet substitution ($GAG \rightarrow GUG$ on mRNA) and 6th amino acid switch (Glutamic acid $\rightarrow$ Valine).
- **Pedigree Conventions:** When drawing pedigree trees, keep generation levels horizontal and label birth order from left to right.